

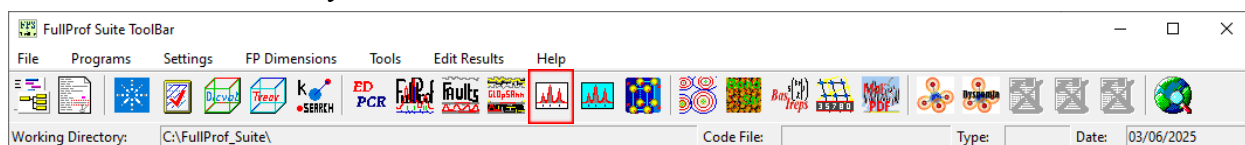
Rietveld Refinement and LeBail Refinement of XRD Data in FullProf Suite

Step 1. Data formatting and background subtraction

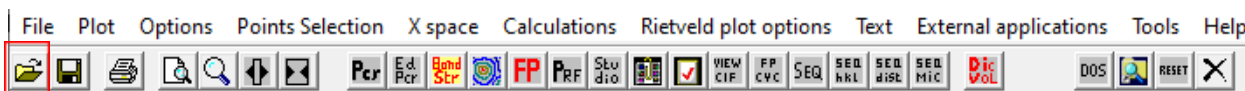
To begin, 2 θ and intensity data must be saved as a .dat file. This can be done by exporting the data from excel or origin as a .txt file, opening this file in notepad, and adding .dat to the end of the filename. The file should be formatted so that the 2 θ values are in the first column and the corresponding intensity value is in the second column (Note: a line can be commented out by inserting an exclamation mark before text):

```
! 2 $\theta$  intensities
5.01675 11075.98296
5.04975 11090.50371
5.08275 11157.33746
5.11575 10944.24285
5.14875 11069.96517
5.18175 11138.82673
5.21475 11121.08617
```

Once you have saved the .dat file to your desired location, open FullProf Suite and click on the WinPlotr tool indicated by the red box:



In the window that opens, click on the option to open a new file:

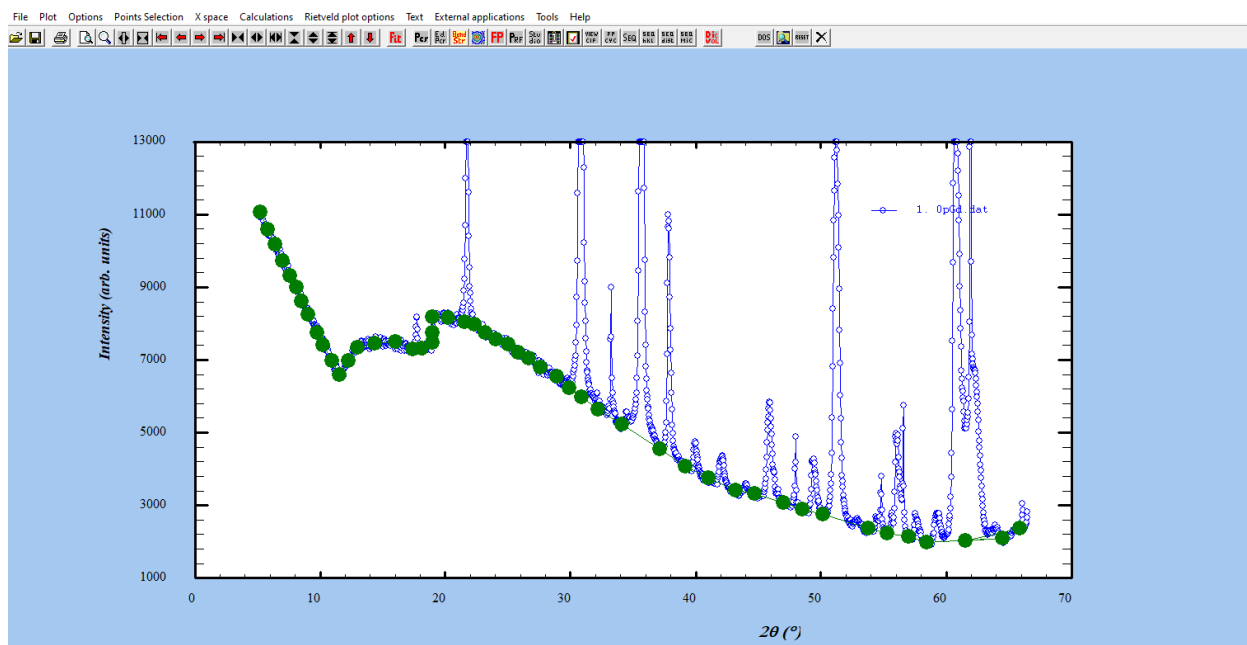


Select the correct format of the file:

Format of data file:

- ☒ 1. X,Y data + INSTRM=10
- ☐ 2. INSTRM=0: Free F.(Ti,step,Tf)
- ☐ 3. INSTRM=1: Old D1A
- ☐ 4. INSTRM=3: D1B (ILL)
- ☐ 5. INSTRM=4: Brookhaven(Synchr.)

After opening the .dat file with WinPlotr, you should see a plot of 2θ vs intensity values. Click **Points Selection > Select background points** and click along the background to add points (Note: drag clicking can be used to enlarge the plot in selected areas, making it easier to select points. Right click to revert to original scale):



Next, navigate to **Points Selection > SaVe background points**. Click “Yes” and save your background points in the same directory as your .dat folder. You should now have the .dat folder containing the XRD data and a .BGR folder containing the background points you have selected. You can now close the WinPlotr tool.

Step 2. Generating a PCR file from a CIF file

Navigate to a website such as Crystallography Open Database

(<https://www.crystallography.net/cod/search.php>) and search for the material that you are

working with. You should see various results with a download link to a CIF file for the given material. After downloading the CIF file, open EdPCR in the main FullProf window:



In the window that appears, click the CIF to PCR tool:



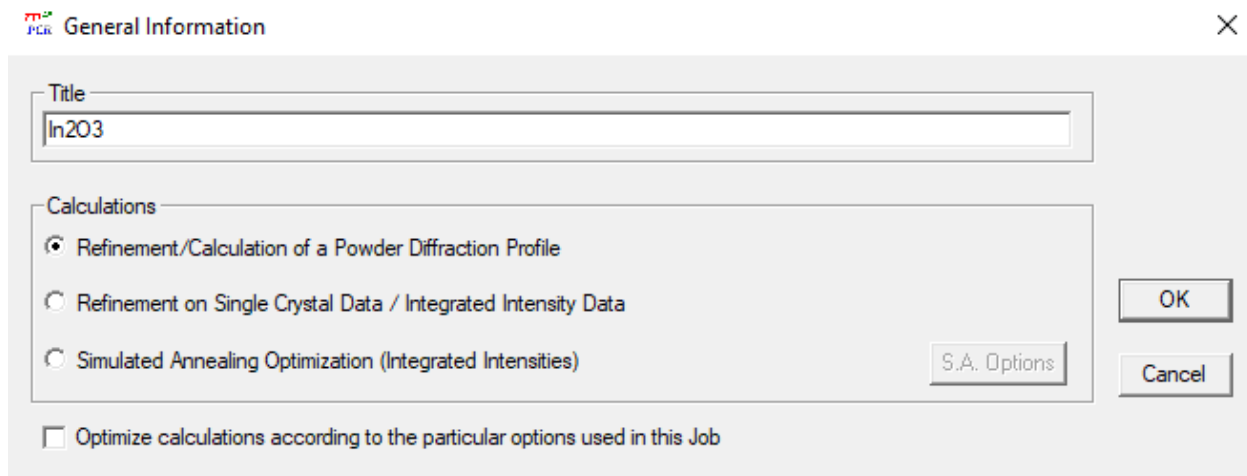
Select the CIF file and click OK in the window that appears.

Step 3. Configuration before refinement

In this section, each underlined header corresponds to a button in the EdPCR menu.

General

In the main EdPCR window, click “General”. Enter an appropriate title and select the calculation option.



Click OK.

Patterns

Next, click on “Patterns” and then “Data file/Peak shape”. In the window that opens, set the data file to your CIF and select the correct format:

Profile Data Information: Pattern 1

Data File / Format | Refinement / Simulation | Pattern Calculation/Peak Shape

Data File: 1010341 Browse...

Format

☐ D1A/D2B (Old Format) ☐ Free Format (2theta, step, 2ThetaF) ☐ Variable Time X-ray Data
☐ D1A/D2B/3T2/G42 ☐ Two Axis Instrument, G41 ☒ X,Y,SIGMA (XYDATA)
☐ D1B (Old Format) ☐ GSAS Format ☐ X'Celerator (PANalytical)
☐ D1B/D20 ☐ Socabim Software ☐ ISIS multi-bank normalized
☐ D4/D20L ☐ Synchrotron (Brookhaven)
☐ DMC/HRPD (P.S.I.) ☐ Synchrotron (DBWS Software)

OK Cancel

Under the “Refinement / Simulation” tab, the settings should be as follows:

Profile Data Information: Pattern 1

Data File / Format | Refinement / Simulation | Pattern Calculation/Peak Shape

Simulation / Refinement Data

☒ X-Ray ☐ Pattern Calculation (X-Ray)
☐ Neutron - CW (Nuclear and Magnetic) ☐ Pattern Calculation (Neutron - CW)
☐ Neutron - T.O.F. (Nuclear and Magnetic) ☐ Pattern Calculation (Neutron - T.O.F.)

Wavelength

Cu λ_1 1.540560 λ_2 1.544390 (I_2 / I_1) 0.5000

OK Cancel

Under the “Pattern Calculation / Peak Shape” tab, the settings should be as follows:

Profile Data Information: Pattern 1

Peak Shape
 Thompson-Cox-Hastings pseudo-Voigt * Axial divergence asymmetry
 ☒ Codefil.SHP
 ☐ Global.SHP

Scattering Variable
☒ 2Theta
 ☐ T.O.F. (microseconds)
 ☐ Energy (keV)

Range
 Theta_min: 5.1817 Theta_max: 64.9777 Step: 0.0330
 Range of calculation of a single reflection in units of FWHM: 8.0000
 Incident beam angle at sample surface (°): 0.000

Theta_min, Theta_max, and Step settings should be adjusted to the specific data. Click “OK” and then click “Background Type”, and make sure this tab has the following configuration:

Background Information

Background Mode
☐ 6-Coefficients polynomial function
☐ 12-Coefficients polynomial function Origin of the polynomial: 40.000
☐ Debye-like (12-coeff.)+ polynomial functions (6-coefficients)
☐ 12-Coefficients Fourier-cosine series
☐ Fourier Filtering Number of points taken for Fourier Filter: 0.0000
☐ Background File transformed by 4-coefficients expression

☒ Linear Interpolation between a set background points with refinable heights
☐ Interpolation by cubic splines
☐ Chebychev Polynomial (24 coefficients)

Click “OK” to go back to the main patterns information menu and then click “OK” once again to navigate back to the main EdPCR menu.

Phases

Click “Phases”. The settings in the main window should be the following:

Phase Information: Phase 1

General Information on Phases

Name of Phase :

Calculation:

Coefficient to calculate the weight percentage of the Phase: ☐ Calculated automatically ☒ Provided by user

Contribution to patterns, preferred orientation direction, reflection list, ...

Space Group symbol/number, symmetry operators, basis functions, etc

Next, click “Contribution to Patterns” and make sure the settings are as follows:

Pattern Contribution Information for Phase 1

Pattern 1 | Pattern 2 | Pattern 3 | Pattern 4 | Pattern 5 | Pattern 6 | Pattern 7

☒ Current Phase contributes to the pattern

Type of Pattern

☒ X-Ray ☐ Pattern Calculation (X-Ray)

☐ Neutron (Constant Wavelength) ☐ Pattern Calculation (Neutron - Constant Wavelength)

☐ Neutron (T.O.F.) ☐ Pattern Calculation (Neutron - T.O.F.)

☐ Nuclear and Magnetic

Peak Shape

Thompson-Cox-Hastings pseudo-Voigt * Axial divergence asymmetry

☒ Codefil.shp ☐ Global.shp

Intensities

Reflection list: Automatically generated from the Space Group symbol

☐ Use special control of parameters for peak overlap, rejected reflections for current phase:

Brindley coefficient: 0.0000

Global weight of the integrated intensity data vs profile data: 0.0000

Factor for excluding reflections [$I < \text{Factor} * \text{Sigma}(I)$]: 0.0000

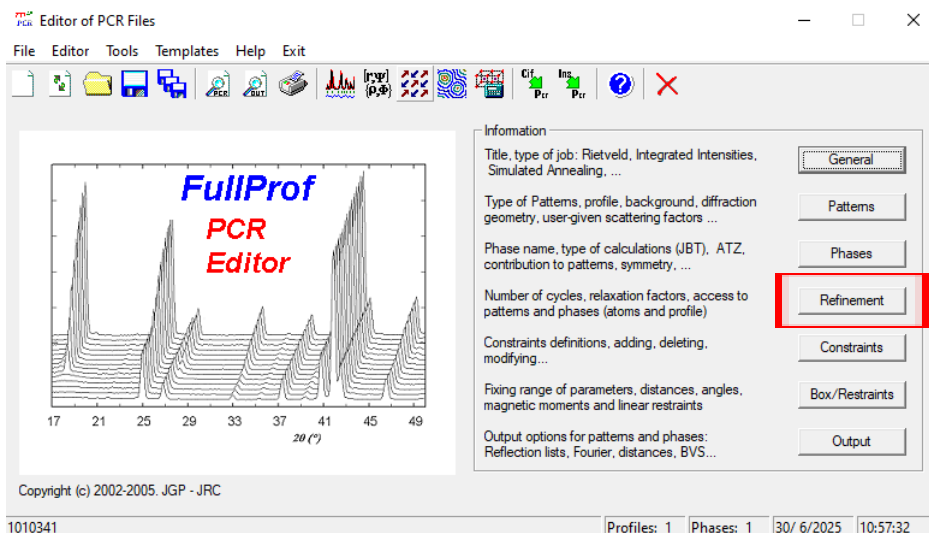
Weights are divided by reduced χ^2 of precedent cycle: 0.0000

OK Cancel

The “Peak Shape” option should be the same setting as in the Patterns section. Click “OK” and then return to the main EdPCR window.

Step 4. Refinement of sample(s)

Click on “Refinement” in the main FullProf window:



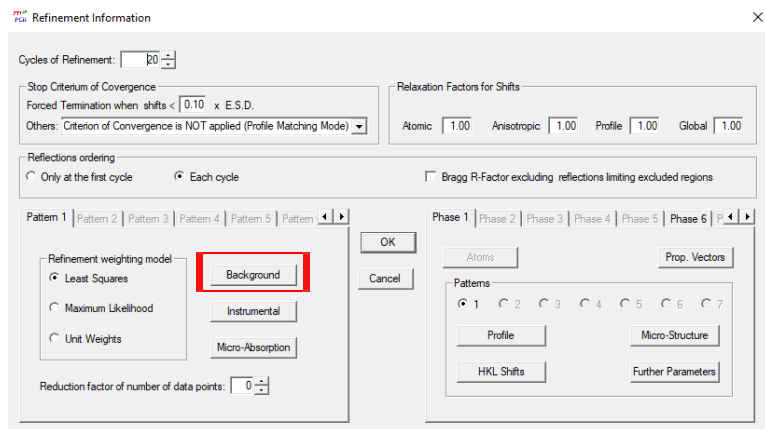
In the window that opens, click “Background” and then “Import from Background File” and select the BGR file that we created earlier:

Interpolation Method

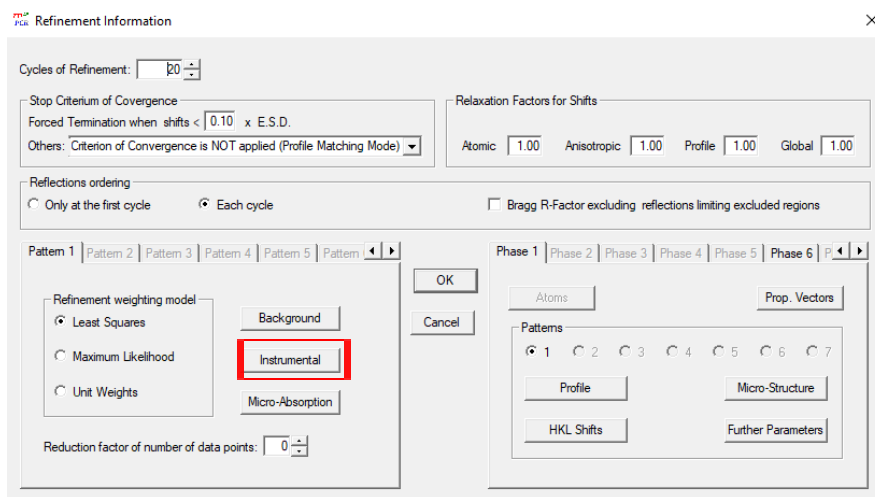
☒ Linear Interpolation ☐ Cubic Splines Interpolation

Information

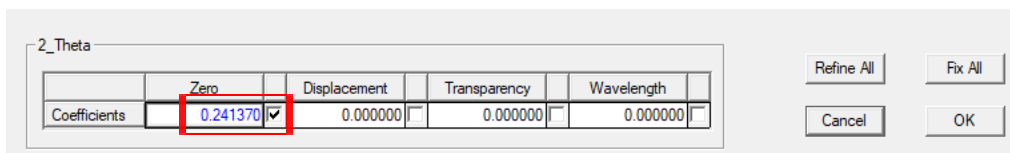
Number of Points: 51



Click “OK” to return to the refinement menu. Next, click “Instrumental” and make sure that the zero coefficient is checked:



Instrumental Parameters Refinement: Pattern 1



Click “OK” to return to the refinement menu.

Click on “Profile”:

Refinement Information

Cycles of Refinement:

Stop Criterion of Coverage
 Forced Termination when shifts < x E.S.D.
 Others: Criterion of Convergence is NOT applied (Profile Matching Mode)

Relaxation Factors for Shifts
 Atomic Anisotropic Profile Global

Reflections ordering
☐ Only at the first cycle ☒ Each cycle ☐ Bragg R-Factor excluding reflections limiting excluded regions

Pattern 1 | Pattern 2 | Pattern 3 | Pattern 4 | Pattern 5 | Pattern 6 |

Refinement weighting model
☒ Least Squares ☐ Maximum Likelihood ☐ Unit Weights

Reduction factor of number of data points:

Phase 1 | Phase 2 | Phase 3 | Phase 4 | Phase 5 | Phase 6 |

Atoms

Patterns
☒ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7
☒ Profile

Make sure that the settings match the following screenshot:

Profile Parameters: Phase 1 Pattern 1

Factors

	Scale	Overall B-factor
Coefficients	0.57313E-04	0.0000

Cell Parameters

	a	b	c	alpha	beta	gamma
Coefficients	☒ 10.138197	☒ 10.138197	☒ 10.138197	90.000	90.000	90.000

FWHM / Shape Parameters | Asymmetry Parameters | Preferred Orientation

FWHM Parameters

	U	V	W	IG
Coefficients	☒ 0.163712	-0.076180	☒ 0.054328	0.000000

Shape Parameters

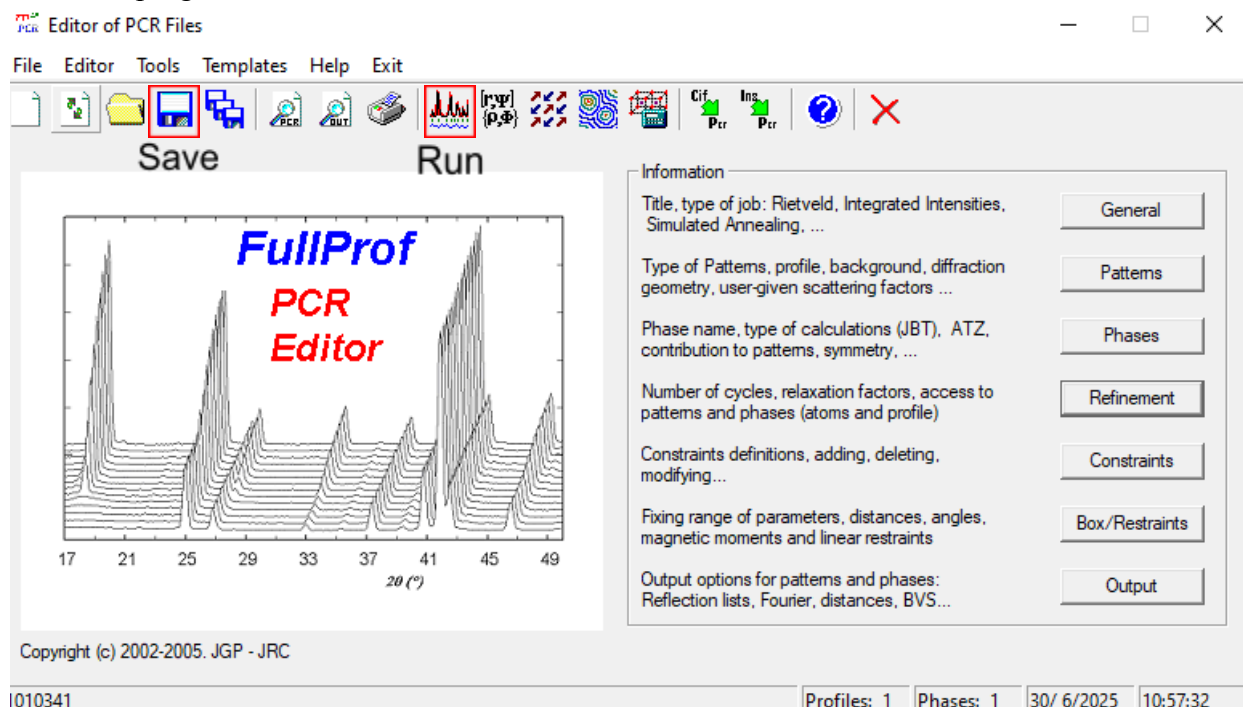
	Eta_0	X
Coefficients	0.000010	0.018961

☐ Refine FWHM for second wavelength

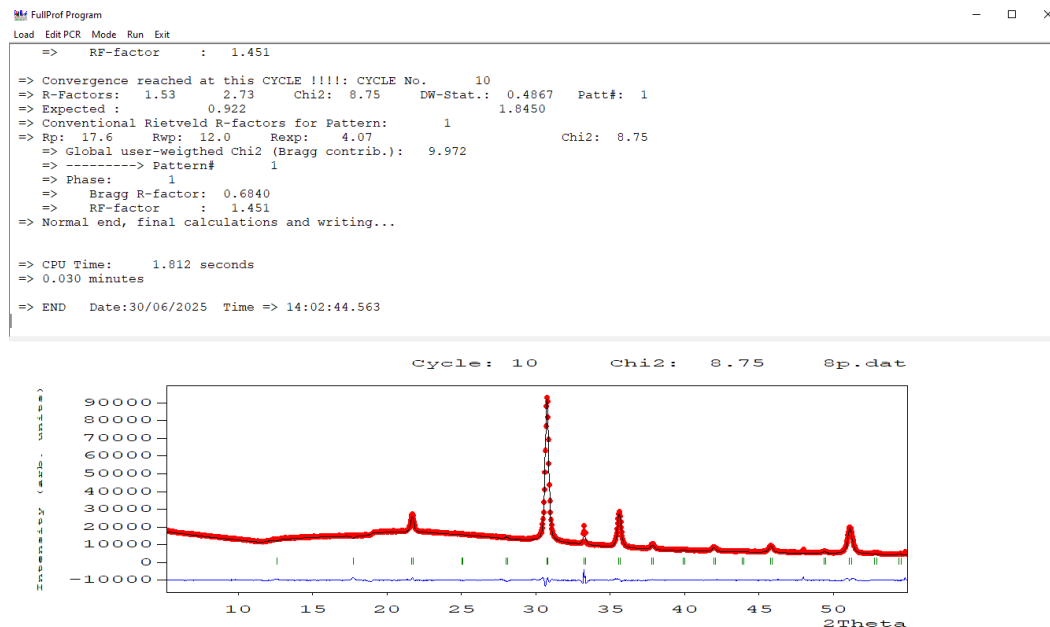
	U2	V2	W2
Coefficients			

Click “OK” to return to the refinement menu and click “OK” once again to return to the main FullProf Suite window.

To begin refining the data, make sure to save changes to the PCR file by clicking the blue save icon in the EdPCR toolbar and then click on the red graph icon indicated in the below screenshot to run the program:



The program should run and your output should look similar to the following:



Close the window and reload the PCR file.

Next, copy and paste all of the files in the working directory into a separate folder. This is so we have a separate directory to refine the data again using the LeBail method.

Name		Date modified	Type	Size
Lebail		6/30/2025 9:13 AM	File folder	
8p	Copy and paste into LeBail folder	6/4/2025 1:34 PM	DAT File	45 KB
8pbq		6/4/2025 2:20 PM	BGR File	3 KB
1010341		6/4/2025 2:21 PM	OUT File	53 KB
1010341		6/4/2025 2:21 PM	PCR File	6 KB
1010341		6/4/2025 2:21 PM	PRF File	98 KB
1010341.sum		6/4/2025 2:21 PM	SUM File	6 KB
10103411.fst		6/4/2025 2:21 PM	FST File	1 KB
edpcr.set		6/4/2025 2:23 PM	SET File	1 KB
fullprof.dir		6/4/2025 2:23 PM	DIR File	1 KB

Click on the PCR file within the LeBail folder that you just copy and pasted into. This should open a new instance of EdPCR. Change the type of refinement by clicking on the “Phases” button and selecting “Profile Matching with constant scale factor” as the option in the calculation drop down menu:

Phase Information: Phase 1

General Information on Phases

Name of Phase : ln203

Calculation: Profile Matching with constant scale factor

Coefficient to calculate the weight percentage of the Phase: 4442.16 ☐ Calculated automatically ☒ Provided by user

Contribution to patterns, preferred orientation direction, reflection list, ... Contribution to Patterns

Space Group symbol/number, symmetry operators, basis functions, etc Symmetry

Initial Previous Add Del Next Last

OK Cancel

Click “OK” to return to the menu and then save and run the PCR file the same as before.

Create another subfolder within the LeBail folder and copy the files once again:

opt	7/1/2025 2:49 PM	File folder	
Op	6/4/2025 1:33 PM	DAT File	45 KB
Opbg	6/4/2025 1:37 PM	BGR File	2 KB
1010341.new	6/30/2025 2:46 PM	NEW File	6 KB
1010341	7/1/2025 2:49 PM	OUT File	67 KB
1010341	7/1/2025 2:49 PM	PCR File	6 KB
1010341	7/1/2025 2:49 PM	PRF File	94 KB
1010341.sum	7/1/2025 2:49 PM	SUM File	5 KB
10103411.fst	6/4/2025 1:39 PM	FST File	1 KB
10103411.hkl	7/1/2025 2:49 PM	HKL File	3 KB
edpcr.set	7/1/2025 2:49 PM	SET File	1 KB
fullprof.dir	7/1/2025 2:49 PM	DIR File	1 KB

Open the PCR file in the new folder and navigate to Patterns > Geometry/IRF > Corrections. Look for the option “Peaks below this 2Theta limit are corrected for asymmetry” and set this option to whatever your max 2Theta value is:

Pattern Diffraction Geometry Information: Pattern 1

Diffraction Geometry

Bragg-Brentano or Debye-Scherrer Geometry

IRF Corrections

External Correction of Integrated/Profile Intensities

☒ None ☐ Linear Interpolation ☐ Empirical Function

Preferred Orientation

☒ Rietveld-Toraya (Exponential Function) ☐ Modified March's Function

☐ March-Dollase Multiaxial Function ☐ March-Dollase Numeric Multiaxial Function

Others

Correction for illuminated area exceeding the Sample surface: 0.000

Monochromator Polarization Correction ($\cos^2 2\theta_{\text{Mon}}$): 0.9100

Fraction of mosaic-crystal (transmission): 0.0000

Absorption Correction Coefficient (μ_R): 0.0000

Peaks below this 2Theta limit are corrected for asymmetry: 52.00

Absorption correction (T.O.F.): None

OK Cancel

Click “OK” and then click “Refinement”. Click “Profile” and under shape parameters, check the X parameter to add this to refinement:

Shape Parameters

	Eta_0	X				
Coefficients	0.000010	0.011774	☒	☐	☐	☐

Save and run the program.

Finding Lattice Parameters and Associated Errors

In the subfolder located in the LeBail folder, there will be an output file with the name of the CIF file used at the beginning. This is what it looks like in my specific case:

Name	Date modified	Type	Size
Op	6/4/2025 1:33 PM	DAT File	45 KB
Opbg	6/4/2025 1:37 PM	BGR File	2 KB
1010341.new	6/30/2025 2:02 PM	NEW File	6 KB
1010341	6/30/2025 2:02 PM	OUT File	120 KB
1010341	6/30/2025 2:02 PM	PCR File	6 KB
1010341	6/30/2025 2:02 PM	PRF File	94 KB
1010341.sum	6/30/2025 2:02 PM	SUM File	8 KB
10103411.fst	6/4/2025 1:39 PM	FST File	1 KB
10103411.hkl	6/30/2025 2:02 PM	HKL File	3 KB
edpcr.set	6/30/2025 2:02 PM	SET File	1 KB
fullprof.dir	6/30/2025 2:02 PM	DIR File	1 KB

Double click on this file to open it in a text editor and search the file for a header that says “Cell Parameters”. The first number will be the lattice constant and the third number the associated error:

=> Cell parameters:

Lattice Constant	11.115504	0.696548	0.264041	Error
	11.115504	0.696548	0.264041	
	11.115504	0.696548	0.264041	
	90.000000	0.000000	0.000000	
	90.000000	0.000000	0.000000	
	90.000000	0.000000	0.000000	